

PROJECT REPORT: BIKE STORE ANALYTICS

End-to-End Data Pipeline & Interactive Business Intelligence Dashboard

1. Executive Summary

This project demonstrates a comprehensive Data Engineering and Analytics workflow. By transforming raw retail datasets into a structured Relational Database Management System (RDBMS) and subsequently developing a Python-based web application, we have enabled real-time business intelligence. The solution provides deep insights into sales performance across brands, categories, and physical store locations.

2. Technical Stack

PostgreSQL

Python 3.14

Streamlit

Pandas

Plotly Express

macOS / VS Code

3. Database Architecture & ETL

The backend is built on **PostgreSQL** using a Star Schema design to optimize query performance and maintain data integrity.

- **Relational Mapping:** Established Primary and Foreign Key constraints across tables including customers, orders, products, and brands.
- **Data Cleaning:** Handled "Ghost Rows" and null-value constraints during the CSV import process to ensure a 100% success rate (4,722 records imported).
- **Financial Logic:** Revenue is calculated dynamically using the formula:

$$\text{Revenue} = \sum (\text{List_Price} \times \text{Quantity} \times (1 - \text{Discount}))$$

4. Data Pipeline Development

A secure bridge was established between the database and the UI using a **Python Virtual Environment (.venv)** on macOS.

Key Pipeline Stages:

1. **Connect:** Secure handshake via psycopg2 library.
2. **Query:** Execution of complex SQL Joins directly from the Python script.
3. **Transform:** Normalization of data into Pandas DataFrames for rapid processing.
4. **Visualize:** Rendering interactive Plotly charts in a Streamlit web container.

5. Core Business Insights

Metric	Key Finding
Top Brand	Trek leads significantly with over \$4.6M in total revenue.
Product Mix	Category-level analysis identified high-volume vs. high-margin items.
Store Performance	Identified regional leaders for targeted marketing and inventory allocation.

6. Conclusion

This project successfully moves beyond static data reporting into the realm of Data Apps. The resulting dashboard is a scalable, interactive tool that empowers decision-makers with live data access, proving the value of a full-stack technical approach to modern data analysis.